KUET · ESE 4025 · HVAC&R System Design

ESE 4025 Question Map

Every question from the 2020, 2021, 2023, 2024 term papers, grouped by topic and ranked by how often it comes back and how many marks it carries.
Answers are in the ESE 4025 Study Pack (HTML and PDF) next to this file.

24-hour list | Pass plan | Frequency grid | Section A | Section B

24-hour priority list

Work top to bottom. Blocks 1 to 4 take about 14 hours, which leaves time for sleep, meals and a final read of the one-shot notes. **Chance** is how often the topic came up, with newer papers counting more (2024 ×4, 2023 ×3, 2021 ×2, 2020 ×1): 80% or more is **Very likely**, 60-79% **Likely**, below that **Possible**. **Level** is how hard it is to learn from zero. Inside each section, the pack below follows this same order.

Exception worth knowing: A4 (the R22 80 kW numerical) shows as only Likely, but the 2021 and 2024 papers asked it word for word for 13-15 marks. If Block 3 goes well, do A4 next. You answer 3 of 4 questions in each section, so Blocks 1-3 already cover enough of both sections to pass.

Block 1: very likely and easy. Quick marks first · about 4.2 h (hours 0.0 to 4.2)

#	Topic	Section	Chance	Marks so far	Level	Time
1	B1 Psychrometric definitions	Section B	Very likely · 100%	52	Easy	25 min
2	B4 Thermal comfort and the ASHRAE comfort chart	Section B	Very likely · 100%	49	Easy	25 min
3	A3 Ton of refrigeration, COP and ice / cold-storage load	Section A	Very likely · 100%	43	Easy	25 min
4	B5 Indoor air quality (IAQ) and ventilation	Section B	Very likely · 100%	43	Easy	25 min

5	A7 Compressors: hermetic sealed, classification	Section A	Very likely · 100%	37	Easy	25 min
6	A6 Refrigerant selection: environmental and safety properties	Section A	Very likely · 100%	33	Easy	25 min
7	A13 Thermoelectric, magnetocaloric and other novel cooling	Section A	Very likely · 90%	35	Easy	25 min
8	A2 Effect of evaporator and condenser temperature on COP	Section A	Very likely · 90%	34	Easy	25 min
9	B2 Psychrometric processes and the air washer	Section B	Very likely · 90%	22	Easy	25 min
10	A1 Vapour compression refrigeration (VCR) cycle	Section A	Very likely · 80%	34	Easy	25 min

Block 2: very likely, medium effort · about 3.8 h (hours 4.2 to 7.9)

#	Topic	Section	Chance	Marks so far	Level	Time
11	A8 Vapour absorption (VAR): single, double, triple effect	Section A	Very likely · 100%	66	Medium	45 min
12	B6 Air-conditioning systems: zone, all-air CAV / VAV, dual duct	Section B	Very likely · 100%	63	Medium	45 min
13	A5 Multistage / multipressure systems and flash cooler	Section A	Very likely · 100%	59	Medium	45 min
14	B9 Duct sizing numerical (equal friction method)	Section B	Very likely · 100%	56	Medium	45 min
15	B7 VRF systems (two-pipe / three-pipe heat recovery)	Section B	Very likely · 100%	47	Medium	45 min

Block 3: very likely numericals (hard but they repeat) · about 2.0 h (hours 7.9 to 9.9)

#	Topic	Section	Chance	Marks so far	Level	Time
16	B8 Cooling load: components, assumptions, numerical	Section B	Very likely · 100%	54	Hard	60 min
17	B3 Summer AC with ventilation air and bypass factor (numerical)	Section B	Very likely · 90%	41	Hard	60 min

Block 4: likely. Backup for your third answer · about 3.8 h (hours 9.9 to 13.7)

#	Topic	Section	Chance	Marks so far	Level	Time
18	A10 Performance of a practical plant: EER, SEER, kW/ton	Section A	Likely · 70%	31	Easy	25 min
19	A15 Cooling towers and condensers	Section A	Likely · 70%	20	Easy	25 min
20	B10 Duct design rules, outlets, dampers, diffusers	Section B	Likely · 60%	35	Easy	25 min
21	A12 Air refrigeration and aircraft cooling	Section A	Likely · 60%	33	Medium	45 min
22	A9 LiBr-water practical problems (crystallisation)	Section A	Likely · 60%	15	Medium	45 min
23	A4 R22 VCR cycle numerical (with table / p-h chart)	Section A	Likely · 70%	40	Hard	60 min

Only if time is left: asked rarely · about 2.9 h

#	Topic	Section	Chance	Marks so far	Level	Time
24	A14 Thermostatic expansion valve	Section A	Possible · 50%	22	Easy	25 min
25	B11 Sol-air temperature, window solar gain, infiltration	Section B	Possible · 50%	19	Medium	45 min
26	A16 Linde air liquefaction	Section A	Possible · 20%	10	Medium	45 min
27	A11 Ammonia ice-plant numerical (COP % of theoretical)	Section A	Possible · 50%	27	Hard	60 min

Pass plan

Each section has 4 questions of 35 marks and you answer 3 (2020 was a shorter 120-mark paper). Section A = refrigeration, Section B = air conditioning. "Must study" topics came up in at least 3 of the 4 papers, or are the same numerical repeated.

SECTION A: REFRIGERATION

SECTION B: AIR CONDITIONING

8 topics

cover 64% of all Section A marks asked

- VCR cycle, COP vs temperatures, TR and ice load
- R22 numerical (same 80 kW question twice)
- Multistage + flash cooler, refrigerants, hermetic compressor
- Absorption: single, double, triple effect

9 topics

cover 89% of all Section B marks asked

- Psychrometric definitions (12 marks every year)
- Summer AC + bypass factor numerical, comfort chart, IAQ
- CAV / VAV / dual duct, VRF three-pipe, cooling load
- Duct sizing numerical (every year, 12 to 15 marks)

- **Never skip a numerical.** R22 cycle, summer AC with bypass factor, and duct sizing together are worth about 45 marks and repeat almost unchanged.
- **Definitions and sketches are cheap marks.** Psychrometric terms, comfort chart, VRF three-pipe, flash cooler and double-effect VAR sketches show up again and again.
- **"Good to have"** topics are your backup for the third question in a section. "Skip if short" appeared once or for few marks.

Frequency grid

Marks asked from each topic in each paper. Darker means more marks. Click a topic to jump to its questions.

Marks per paper: 1–8 9–16 17–24 25+

SECTION A: REFRIGERATION: MARKS ASKED PER PAPER

Topic	2024	2023	2021	2020	Years	Marks	
A3 Ton of refrigeration, COP and ice / cold-storage load	10	7	8	18	4/4	43	Must study
A7 Compressors: hermetic sealed, classification	12	8	10	7	4/4	37	Must study
A6 Refrigerant selection: environmental and safety properties	10	8	8	7	4/4	33	Must study

A13 Thermoelectric, magnetocaloric and other novel cooling	20	10	5		3/4	35	Good to have
A2 Effect of evaporator and condenser temperature on COP	12	10	12		3/4	34	Must study
A1 Vapour compression refrigeration (VCR) cycle	12	10		12	3/4	34	Must study
A8 Vapour absorption (VAR): single, double, triple effect	21	19	16	10	4/4	66	Must study
A5 Multistage / multipressure systems and flash cooler	13	23	12	11	4/4	59	Must study
A10 Performance of a practical plant: EER, SEER, kW/ton	23	8			2/4	31	Good to have
A15 Cooling towers and condensers	6	14			2/4	20	Skip if short
A12 Air refrigeration and aircraft cooling		10	10	13	3/4	33	Good to have
A9 LiBr-water practical problems (crystallisation)	8		7		2/4	15	Good to have
A4 R22 VCR cycle numerical (with table / p-h chart)	13		15	12	3/4	40	Must study
A14 Thermostatic expansion valve		10	12		2/4	22	Good to have
A16 Linde air liquefaction			10		1/4	10	Skip if short
A11 Ammonia ice-plant numerical (COP % of theoretical)		12	15		2/4	27	Good to have

SECTION B: AIR CONDITIONING: MARKS ASKED PER PAPER

Topic	2024	2023	2021	2020	Years	Marks	
B1 Psychrometric definitions	12	18	12	10	4/4	52	Must study
B4 Thermal comfort and the ASHRAE comfort chart	10	11	8	20	4/4	49	Must study
B5 Indoor air quality (IAQ) and ventilation	15	12	10	6	4/4	43	Must study

B2 Psychrometric processes and the air washer	8	6	8		3/4	22	Must study
B6 Air-conditioning systems: zone, all-air CAV / VAV, dual duct	18	12	17	16	4/4	63	Must study
B9 Duct sizing numerical (equal friction method)	15	15	14	12	4/4	56	Must study
B7 VRF systems (two-pipe / three-pipe heat recovery)	12	7	20	8	4/4	47	Must study
B8 Cooling load: components, assumptions, numerical	15	22	7	10	4/4	54	Must study
B3 Summer AC with ventilation air and bypass factor (numerical)	15	11	15		3/4	41	Must study
B10 Duct design rules, outlets, dampers, diffusers		13	14	8	3/4	35	Good to have
B11 Sol-air temperature, window solar gain, infiltration		4	15		2/4	19	Good to have

Section A: Refrigeration questions 1-4

A3 Ton of refrigeration, COP and ice / cold-storage load

Must study · Very likely · 100% · Easy

Asked in 4 of 4 papers · 43 marks in total

STUDY THESE

- 1 TR = 3.517 kW (210 kJ/min): melting 1 ton of ice in 24 h
- COP; refrigerator vs heat pump; refrigeration vs air conditioning
- Load = m(cΔT + latent) / time, then divide by 3.517 for TR

> 2024 Q1(a) 10 marks
>
> Define Ton. Calculate the refrigeration tonnage for 2000 kg ice melting from 0 °C to 45 °C.

> 2023 Q1(d) 7 marks
>
> Calculate the refrigeration tonnage for 1500 kg ice from 0 °C to 25 °C.

> 2021 Q1(a) 8 marks
>
> Define Ton and COP of refrigeration system. What is the basic difference between a refrigerator and a heat pump?

> 2020 Q1(a) 8 marks
>
> What is meant by TR? Write down the differences between refrigeration and air conditioning system.

> 2020 Q1(c) 10 marks
>
> 600 kg of fruits are supplied to a cold storage at 25 °C. The cold storage is maintained at −5 °C and the fruits get cooled to the storage temperature in 8 hours. The latent heat of freezing is 105 kJ/kg and specific heat of fruit is 1.255 kJ/kgK. Find the refrigeration capacity of the plant.

Full answer: ESE 4025 Study Pack, topic A3.

A7 Compressors: hermetic sealed, classification

Must study · Very likely · 100% · Easy

Asked in 4 of 4 papers · 37 marks in total

STUDY THESE

- Hermetic: motor and compressor in one welded shell
- Advantages / disadvantages list (memorise 5 each)
- Compressor classification; centrifugal vs reciprocating

> 2024 Q3(d) 5 marks
>
> Write down the advantages and disadvantages of hermetic sealed compressor.

> 2024 Q3(c) 7 marks
>
> Classify compressors. Write down the advantages of centrifugal compressors over reciprocating compressors.

> 2023 Q3(c) 8 marks
>
> What is meant by hermetic sealed compressor? Write down its advantages and disadvantages.

> 2021 Q3(d) 10 marks
>
> What is meant by hermetic sealed compressor? What are the advantages and disadvantages of this compressor?

> 2020 Q3(b) 7 marks
>
> What is meant by hermetic sealed compressor? What are its advantages?

Full answer: ESE 4025 Study Pack, topic A7.

A6 Refrigerant selection: environmental and safety properties

Must study · Very likely · 100% · Easy

Asked in 4 of 4 papers · 33 marks in total

STUDY THESE

- Thermodynamic, chemical, physical desirable properties (short list)
- Environmental: ODP, GWP, atmospheric life
- Safety: toxicity, flammability (ASHRAE A1…B3 class)

> 2024 Q3(b) 10 marks
>
> What are the major criteria of refrigerant selection? Explain the environmental and safety properties of a good refrigerant.

> 2023 Q3(b) 8 marks
>
> Discuss the environmental and safety properties to consider before selecting a refrigerant.

> 2021 Q2(b) 8 marks
>
> Briefly discuss the desirable characteristics of a fluid to be used as a refrigerant.

> 2020 Q2(a) 7 marks
>
> Explain the environmental and safety properties of refrigerant.

Full answer: ESE 4025 Study Pack, topic A6.

A13 Thermoelectric, magnetocaloric and other novel cooling

Good to have · Very likely · 90% · Easy

Asked in 3 of 4 papers · 35 marks in total

STUDY THESE

- Peltier effect; reversing current swaps hot and cold sides
- TE vs solar-assisted vs electro-caloric (compare COP)
- Why magnetocaloric is not commercial yet (magnets, cost, small ΔT)

> 2024 Q8(a) 10 marks

> Explain the working principle of the thermoelectric cooling technique. How does reversing current affect its operation? (asked in Section B)

> 2024 Q8(b) 10 marks
>
> Explain the reason why magneto-caloric cooling has not yet achieved commercial adoption in building HVAC system. (Section B)

> 2023 Q4(d) 10 marks
>
> How thermo-electric cooling technique differs from solar-assisted cooling system? Explain which one is more efficient system and why?

> 2021 Q3(b) 5 marks
>
> Differentiate between thermo-electric and electro-caloric cooling systems.

Full answer: ESE 4025 Study Pack, topic A13.

A2 Effect of evaporator and condenser temperature on COP

Must study · Very likely · 90% · Easy

Asked in 3 of 4 papers · 34 marks in total

STUDY THESE

- Lower evaporator temp: less refrigerating effect, more work, lower COP (p-h sketch)
- Higher condenser temp: same trend (p-h sketch)
- Evaporator temperature has the stronger influence

> 2024 Q2(a) 12 marks
>
> Discuss the effect of evaporating and condensing temperature on the cycle efficiency with necessary diagrams.

> 2023 Q1(b) 10 marks
>
> Explain the effects of evaporating and condensing temperature on the performance of vapor compression refrigeration cycle.

> 2021 Q1(b) 12 marks
>
> How does the increase in condenser temperature affect on COP? Also briefly explain the influence of evaporator temperature on COP. Which of the two temperatures have more influence on COP?

Full answer: ESE 4025 Study Pack, topic A2.

A1 Vapour compression refrigeration (VCR) cycle

Must study · Very likely · 80% · Easy

Asked in 3 of 4 papers · 34 marks in total

STUDY THESE

- Four components and four processes (compression, condensation, throttling, evaporation)
- T-s and p-h diagrams of the ideal cycle, with state points 1-2-3-4
- Refrigerating effect, work, COP formulas from enthalpies

> 2024 Q1(b) 12 marks
>
> Explain thermodynamic process of an ideal vapor compression refrigeration (VCR) cycle with necessary sketches.

> 2023 Q1(a) 10 marks
>
> Describe the working principle of vapor compression refrigeration system with neat sketch.

> 2020 Q1(b) 12 marks
>
> Discuss about the thermodynamic process of theoretical vapor compression refrigeration cycle with dry saturated vapor after compression with necessary diagrams.

Full answer: ESE 4025 Study Pack, topic A1.

A8 Vapour absorption (VAR): single, double, triple effect

Must study · Very likely · 100% · Medium

Asked in 4 of 4 papers · 66 marks in total

STUDY THESE

- Single-effect LiBr-water schematic (generator, absorber, condenser, evaporator, HX)
- Double effect: two generators, higher COP (~1.2); triple effect (~1.7)
- Classification of VAR; VAR vs VCR; advantages of VAR; heat sources

> 2024 Q4(b) 13 marks
>
> Describe the commercial double-effect VAR system with necessary sketch.

> 2024 Q4(c) 8 marks

> Briefly explain the different heat sources of VAR system.

> 2023 Q3(a) 13 marks
> Write down the classification of vapor absorption refrigeration system. Describe the triple effect vapor absorption system with a schematic diagram.

> 2023 Q3(d) 6 marks
> Differentiate between VCR and VAR systems.

> 2021 Q4(d) 10 marks
> Draw the schematic of a single effect absorption refrigeration system. Also explain how it is different from double effect absorption refrigeration system.

> 2021 Q4(b) 6 marks
> What are the advantages of vapor absorption system over vapor compression refrigeration system?

> 2020 Q3(a) 10 marks
> Draw double effect absorption refrigeration system and explain how it is different from triple effect absorption refrigeration system.

Full answer: ESE 4025 Study Pack, topic A8.

A5 Multistage / multipressure systems and flash cooler

Must study · Very likely · 100% · Medium

Asked in 4 of 4 papers · 59 marks in total

STUDY THESE

- Why and when: large pressure ratio, high discharge temp, low volumetric efficiency
- Two-stage expansion with flash chamber (schematic + p-h)
- Single compressor & multiple expansion valves; individual compressors & multiple expansion valves
- Advantages and disadvantages of multistage

> 2024 Q3(a) 13 marks
> Why and when multi-stage VCR system is required? Explain two-stage expansion system with a flash cooler with necessary sketch.

> 2023 Q2(a) 10 marks
> Why multi-stage vapor compression is required? Write down the advantages and disadvantages of a multi-stage vapor compression system.

> 2023 Q2(b) 13 marks
>
> Describe two-stage expansion system with a flash cooler with necessary diagrams.

> 2021 Q2(a) 12 marks
>
> What is meant by multipressure systems? Discuss about the individual compressors-multiple expansion valves with the required diagrams.

> 2020 Q2(b) 11 marks
>
> Define multistage compression. Explain the single compressor-multiple expansion valves with the required p-h diagram.

Full answer: ESE 4025 Study Pack, topic A5.

A10 Performance of a practical plant: EER, SEER, kW/ton

Good to have · Likely · 70% · Easy

Asked in 2 of 4 papers · 31 marks in total

STUDY THESE

- Full-load vs part-load (IPLV) measurement
- EER, SEER, kW/ton, Energy Star, HSPF one-line definitions

> 2024 Q2(b) 8 marks
>
> How the full-load and part-load performance of a practical refrigeration plant measured? Briefly explain.

> 2024 Q2(c) 15 marks
>
> Write notes on: (i) EER, (ii) SEER, (iii) kW/ton, (iv) Energy Star, (v) HSPF.

> 2023 Q1(c) 8 marks
>
> How the performance of a practical refrigeration is evaluated? Describe.

Full answer: ESE 4025 Study Pack, topic A10.

A15 Cooling towers and condensers

Skip if short · Likely · 70% · Easy

Asked in 2 of 4 papers · 20 marks in total

STUDY THESE

- Classification of cooling towers (natural/mechanical draft, counter/cross flow)

- Range, approach, effectiveness
- Air-cooled vs water-cooled condenser

> 2024 Q4(d) 6 marks
>
> Write down the classifications of cooling towers.

> 2023 Q8(c) 9 marks
>
> Write down the classifications of cooling towers? Also describe how the performance of cooling tower is evaluated? (Section B)

> 2023 Q4(c) 5 marks
>
> Write down the differences between air cooled condenser and water cooled condenser.

Full answer: ESE 4025 Study Pack, topic A15.

A12 Air refrigeration and aircraft cooling

Good to have · Likely · 60% · Medium

Asked in 3 of 4 papers · 33 marks in total

STUDY THESE

- Bell-Coleman / simple air cycle with schematic and T-s
- Why aircraft need cooling; methods of air refrigeration
- Aircraft numerical: $Q = m\, c_p(T_{cabin} - T_{turbine\ exit})$

> 2023 Q4(a) 10 marks
>
> Describe the working principle of an air refrigerator system. Also mention the necessity of cooling for an aircraft.

> 2021 Q3(a) 10 marks
>
> What are the different methods of air refrigeration systems? Explain the simple air cooling system with necessary diagrams.

> 2020 Q3(c) 13 marks
>
> In aircraft refrigeration unit using air cycle, 50 kg/min of air at 200 cm Hg gauge and 205 °C are bled off the air compressor serving the jet engine. The bled air passes through a heat exchanger leaving at 180 cm Hg gauge and 80 °C. It is then expanded through a small cooling turbine to 20 cm Hg vacuum and −10 °C. The air exhausted out of the plane is at 25 °C. Assume c_p = 1.0 kJ/kgK. Determine the cooling capacity in TR, and COP based on input power to bled-off air.

Full answer: ESE 4025 Study Pack, topic A12.

A9 LiBr-water practical problems (crystallisation)

Good to have · Likely · 60% · Medium

Asked in 2 of 4 papers · 15 marks in total

STUDY THESE

- Crystallisation, air leakage / vacuum, corrosion, pressure equalisation, solidification of water
- How crystallisation happens and is prevented

> 2024 Q4(a) 8 marks
>
> Briefly explain the various practical problems in water-lithium bromide vapor absorption refrigeration (VAR) system.

> 2021 Q4(c) 7 marks
>
> What are the practical problems in LiBr-water vapor absorption system? Write a short note on crystallization problem.

Full answer: ESE 4025 Study Pack, topic A9.

A4 R22 VCR cycle numerical (with table / p-h chart)

Must study · Likely · 70% · Hard

Asked in 3 of 4 papers · 40 marks in total

STUDY THESE

- Read h1, h2, h3, v1 from R22 table or p-h chart; apply subcooling and superheat
- m = Q / (h1 − h4); volume flow = m × v1; W = m(h2 − h1); COP = (h1 − h4)/(h2 − h1)
- The 80 kW problem came twice word for word

> 2024 Q1(c) 13 marks
>
> A refrigeration system using R22 is to have a refrigerating capacity of 80 kW. The system follows VCR cycle with 5 °C subcooling and 10 °C superheating in which the evaporating and condensing temperature are −10 °C and 45 °C, respectively. Determine: (i) the volume flow rate of refrigerant, (ii) COP, (iii) the power required to drive the compressor.

> 2021 Q1(c) 15 marks

Same question as 2024 1(c): R22, 80 kW, 5 °C subcooling, 10 °C superheating, evaporating −10 °C, condensing 45 °C. Find volume flow rate, COP and compressor power.

`2020` `Q2(c)` `12 marks`

A standard VCR system developing 50 kW of refrigeration using R22 refrigerant operates with a condensing temperature of 35 °C and an evaporating temperature of −10 °C. The enthalpy of the refrigerant after the compression is 435.2 kJ/kg. Using the table, determine: (i) the refrigeration effect in kJ/kg; (ii) the circulation rate of refrigerant in kg/s; (iii) the power required by the compressor in kW; and (iv) COP.

Table: Properties of saturated liquid and saturated vapor of R22

Temperature (°C)	Pressure (bar)	Enthalpy (kJ/kg)		Entropy (kJ/kgK)	
		Liquid	Vapor	Liquid	Vapor
−12	3.3040	186.09	400.27	0.9484	1.7686
−8	3.8059	190.69	401.85	0.9657	1.7621
34	13.215	241.84	414.79	1.1419	1.7050
36	13.898	244.44	415.18	1.1501	1.7024

Full answer: ESE 4025 Study Pack, topic A4.

A14 Thermostatic expansion valve

Good to have · Possible · 50% · Easy

Asked in 2 of 4 papers · 22 marks in total

STUDY THESE

- Function of expansion valve
- TEV sketch: bulb, diaphragm, spring; superheat control

`2023` `Q4(b)` `10 marks`

Explain the thermostatic expansion valve with a neat sketch.

`2021` `Q4(a)` `12 marks`

What is the function of expansion valve in a refrigeration system? Explain the working principle of thermostatic expansion valve.

Full answer: ESE 4025 Study Pack, topic A14.

A16 Linde air liquefaction

Skip if short · Possible · 20% · Medium

Asked in 1 of 4 papers · 10 marks in total

STUDY THESE

- Linde cycle schematic and T-s; Joule-Thomson cooling

> 2021 Q3(c) 10 marks
>
> Describe the Linde system of liquefaction of air.

Full answer: ESE 4025 Study Pack, topic A16.

A11 Ammonia ice-plant numerical (COP % of theoretical)

Good to have · Possible · 50% · Hard

Asked in 2 of 4 papers · 27 marks in total

STUDY THESE

- Q = ice mass × 335 / 86400; Carnot COP = $T_L/(T_H - T_L)$ or from NH_3 table
- Actual COP = % × theoretical; W = Q / COP

> 2023 Q2(c) 12 marks
>
> An ammonia refrigerator produces 25 tonnes of ice from and at 0 °C in 24 hours. The temperature range of the compressor is from 15 °C to −5 °C. The vapor is dry saturated at the end of compression and an expansion valve is used. Assume a COP of 45% of the theoretical value. Calculate the power required to drive the compressor. Latent heat of ice is 335 kJ/kg.

> 2021 Q2(c) 15 marks
>
> Same as above with 30 tonnes of ice, compressor range 20 °C to −10 °C, COP = 55% of theoretical.

Full answer: ESE 4025 Study Pack, topic A11.

Section B: Air conditioning questions 5-8

B1 Psychrometric definitions

Must study · Very likely · 100% · Easy

Asked in 4 of 4 papers · 52 marks in total

STUDY THESE

- 12 marks every single year: four terms, 3 marks each
- SHF (3 times), humid specific heat, degree of saturation, bypass factor, WBT (2 each), specific humidity, DPT, RH
- SHF line on the chart

> 2024 Q5(a) 12 marks
>
> Define: (i) Humid specific heat, (ii) Sensible heat factor, (iii) Degree of saturation, (iv) Bypass factor.

> 2023 Q5(a) 12 marks
>
> Define: (i) Specific humidity, (ii) Degree of saturation, (iii) Wet bulb temperature, (iv) Humid specific heat.

> 2023 Q5(b) 6 marks
>
> What is meant by SHF? How SHF line is drawn in psychrometric chart?

> 2021 Q5(a) 12 marks
>
> Define: (i) Dew point temperature, (ii) Relative humidity, (iii) Wet bulb temperature, (iv) Sensible heat factor.

> 2020 Q4(a) 10 marks
>
> What is meant by bypass factor? How do you draw sensible heat factor (SHF) line on a psychrometric chart?

Full answer: ESE 4025 Study Pack, topic B1.

B4 Thermal comfort and the ASHRAE comfort chart

Must study · Very likely · 100% · Easy

Asked in 4 of 4 papers · 49 marks in total

STUDY THESE

- Draw the ASHRAE comfort chart (5 marks, 3 of 4 years)
- Factors: air temp, MRT, humidity, air speed, clothing, activity
- PMV / PPD (min PPD 5%), effective vs operative temperature, indoor/outdoor design conditions

> 2024 Q6(a) 5 marks
>
> Draw the ASHRAE comfort chart.

> 2024 Q6(b) 5 marks

> Is it impossible to satisfy a thermal environment that satisfies everyone? Explain with the equation: $PPD = 100 - 95 \times e^{-0.03353\,PMV^4 - 0.2179\,PMV^2}$

> `2023` `Q6(a)` `5 marks`
>
> Draw the ASHRAE comfort chart.

> `2023` `Q6(b)` `6 marks`
>
> What is meant by indoor design conditions and outdoor design conditions?

> `2021` `Q6(a)` `8 marks`
>
> What is meant by thermal comfort? What are the factors affecting thermal comfort?

> `2020` `Q4(b)` `10 marks`
>
> What is meant by thermal comfort? What are the factors affecting thermal comfort?

> `2020` `Q4(c)` `6 marks`
>
> Distinguish between effective temperature and operative temperature.

> `2020` `Q4(d)` `4 marks`
>
> Draw the ASHRAE comfort chart.

Full answer: ESE 4025 Study Pack, topic B4.

B5 Indoor air quality (IAQ) and ventilation

Must study · Very likely · 100% · Easy

Asked in 4 of 4 papers · 43 marks in total

STUDY THESE

- Effects of poor IAQ on health and comfort (SBS, CO_2, VOC, particulates)
- Control: source control, ventilation (dilution), filtration
- CO_2 balance: $Q = G / (C_{in} - C_{out})$

> `2024` `Q6(c)` `15 marks`
>
> Dhaka has twice the CO_2 concentration of Khulna (400 ppm). If a class room with 30 students is in a room of same size in both cities, determine the required ventilation rate for each location. Also, explain how to control IAQ in a classroom.

> `2023` `Q6(d)` `12 marks`
>
> What are effects of IAQ on health and comfort? How to control IAQ?

> 2021 Q6(d) 10 marks
>
> Write short notes on: (i) Filtration, and (ii) Dilution.

> 2020 Q5(c) 6 marks
>
> What are the effects of indoor air quality on health and comfort? How to control indoor air quality?

Full answer: ESE 4025 Study Pack, topic B5.

B2 Psychrometric processes and the air washer

Must study · Very likely · 90% · Easy

Asked in 3 of 4 papers · 22 marks in total

STUDY THESE

- Cooling & dehumidification and heating & humidification on the chart
- Air washer: spray chamber, eliminators; which process at which water temperature

> 2024 Q5(b) 8 marks
>
> What is an air washer? Describe the following process with an air washer: (i) Cooling and Dehumidification, (ii) Heating and Humidification.

> 2023 Q5(c) 6 marks
>
> Explain the working principle of an air washer.

> 2021 Q5(b) 8 marks
>
> What is psychrometry? Describe the following process in a psychrometric chart: (i) Cooling and dehumidification, (ii) Heating and humidification.

Full answer: ESE 4025 Study Pack, topic B2.

B6 Air-conditioning systems: zone, all-air CAV / VAV, dual duct

Must study · Very likely · 100% · Medium

Asked in 4 of 4 papers · 63 marks in total

STUDY THESE

- Zone definition; classification of AC systems
- All-air systems: single duct CAV, VAV, dual duct, multizone (schematics)
- CAV vs VAV; advantages of all-air; split AC; rooftop unit pros/cons; system choice matrix

2024 Q7(a) 8 marks

What is meant by zone? Explain dual duct constant air volume air conditioning system with schematic diagram.

2024 Q6(d) 10 marks

Using the data given in the following table, make a system choice matrix and explain which system will be most appropriate for KUET academic building's thermal load.

Criteria	Cooling capacity	Temperature control	Zone timing	Initial cost	Operating maintenance	Energy consumption
Relative importance	08	09	10	05	09	08
Split-type performance	08	08	07	06	06	07
Rooftop unit performance	10	09	08	08	09	09

2023 Q6(c) 12 marks

Describe the various all-air central air-conditioning system.

2021 Q6(b) 9 marks

What is meant by zone? Explain the difference between constant air volume and variable air volume air-conditioning system with schematic diagram.

2021 Q6(c) 8 marks

What are the advantages and disadvantages of rooftop air-conditioning unit?

2020 Q5(a) 9 marks

Classify air-conditioning system. Explain the working principle of split air-conditioning system with schematic diagram.

2020 Q5(b) 7 marks

Distinguish between constant air volume and variable air volume of all-air central air-conditioning system. What are the advantages of all-air systems?

Full answer: ESE 4025 Study Pack, topic B6.

B9 Duct sizing numerical (equal friction method)

Must study · Very likely · 100% · Medium

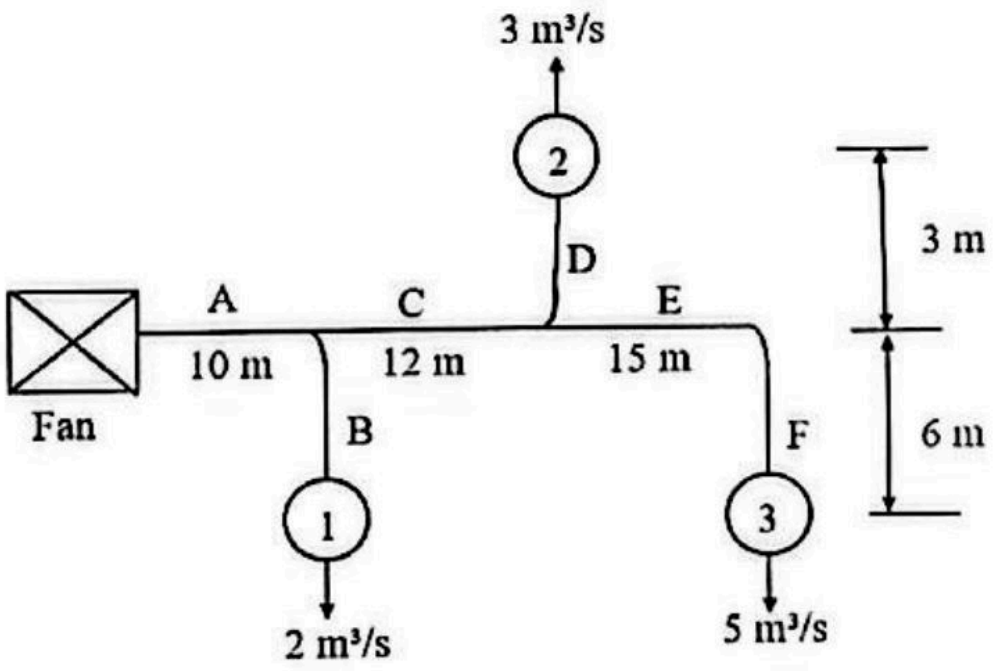
Asked in 4 of 4 papers · 56 marks in total

STUDY THESE

- Asked every year for 12 to 15 marks
- Use the longest run, fixed friction rate; size each section from its flow
- Round D → rectangular a × b with aspect ratio 3:1: $D_{eq} = 1.3(ab)^{0.625}/(a+b)^{0.25}$

2024 Q8(c) 15 marks

For the duct system as shown in figure, determine the dimension of rectangular duct with aspect ratio 3:1. Assume all dynamic loss coefficients are 0.8.



2023 Q8(d) 15 marks

Determine the size of the duct system shown in figure using equal friction method assuming pressure drop of 0.1 mm H_2O/m length.

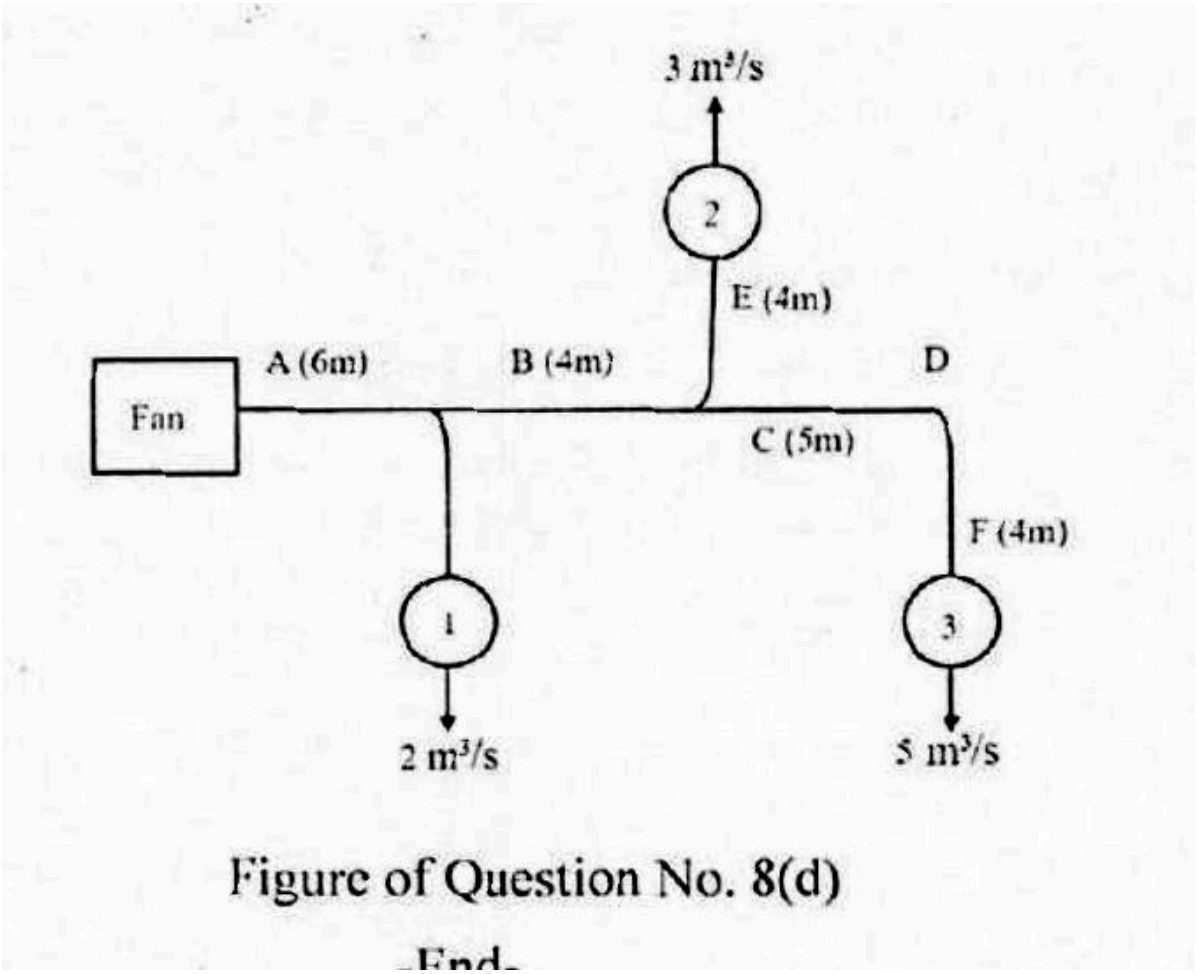
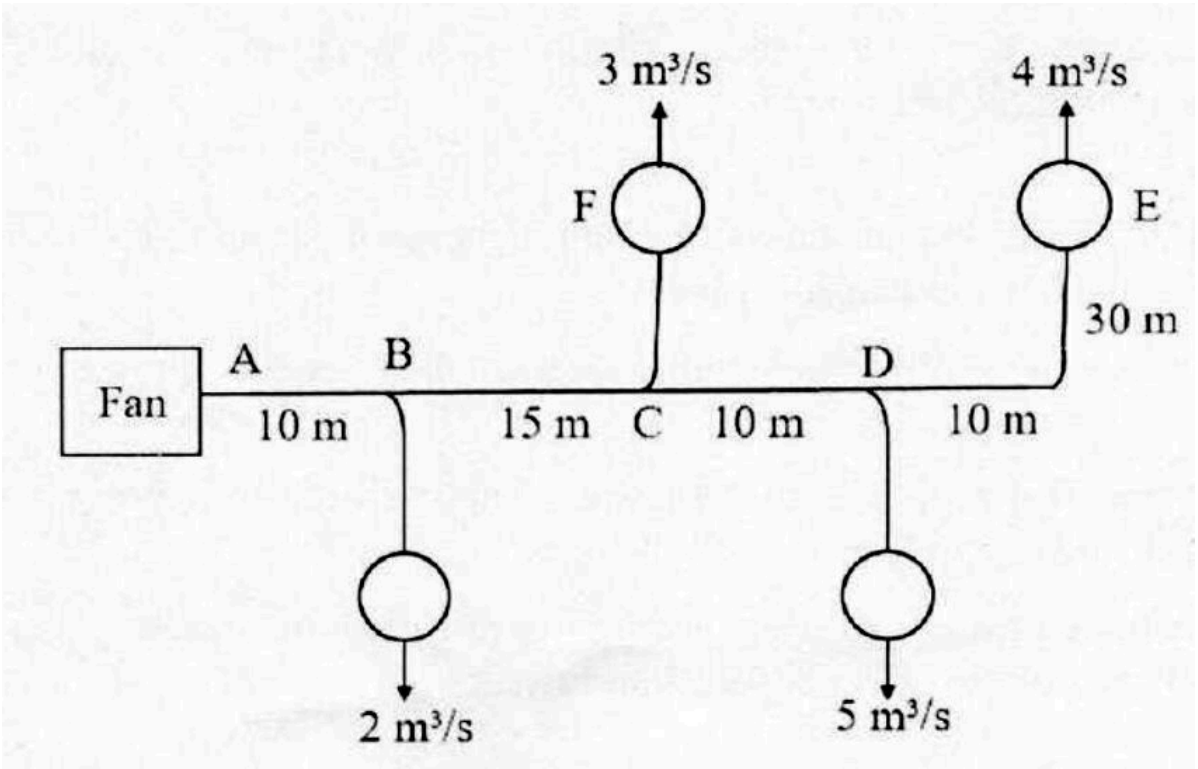


Figure of Question No. 8(d)

-End-

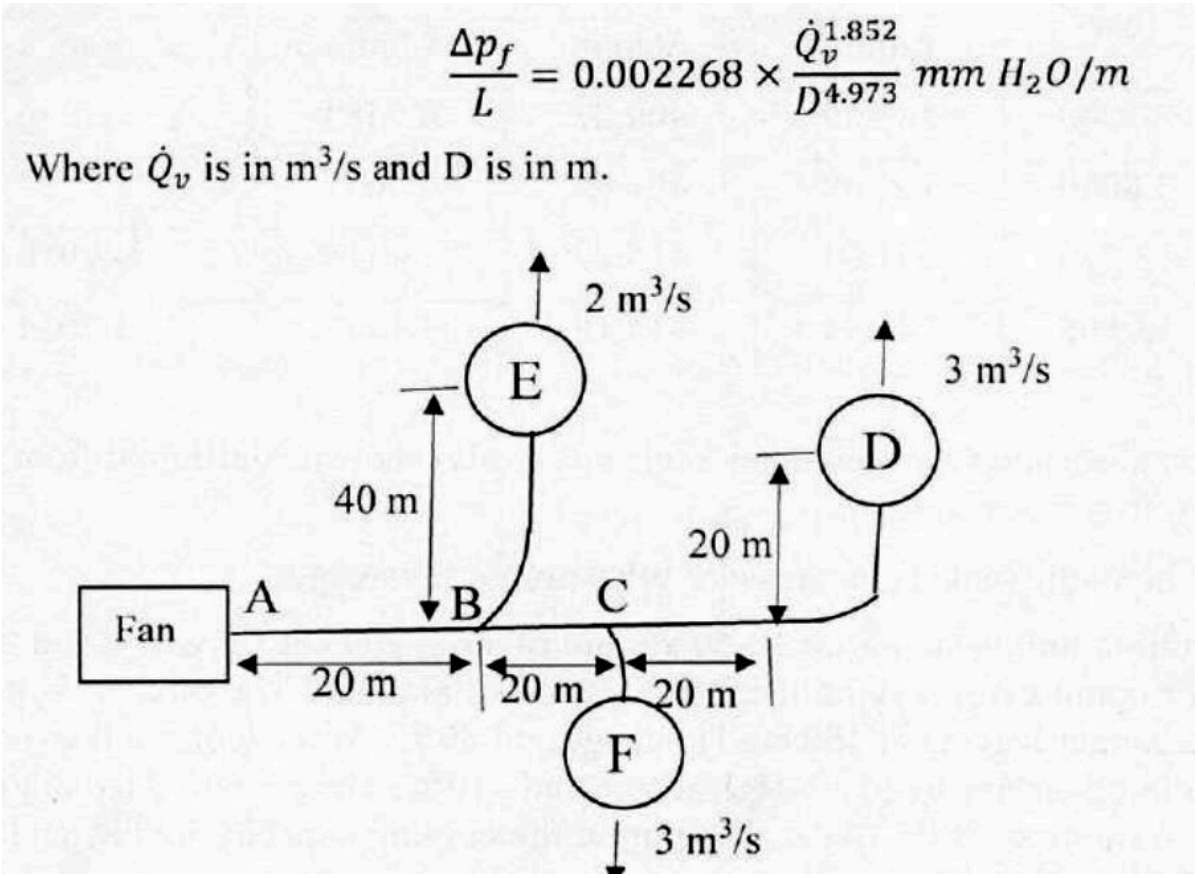
2021 Q8(d) 14 marks

For the duct system as shown in figure, determine the dimensions AB, BC, CD and DE of the rectangular ducts with aspect ratio 3:1 assuming friction rate of 0.1 mm H_2O/m length duct.



2020 Q6(c) 12 marks

Determine the dimensions of AB, BC, CD as shown in figure using equal friction method. Choose a friction rate of 0.09 mm H_2O/m. Use $\Delta p_f/L = 0.002268 \times Q^{1.852}/D^{4.973}$ mm H_2O/m (Q in m^3/s, D in m).



Full answer: ESE 4025 Study Pack, topic B9.

B7 VRF systems (two-pipe / three-pipe heat recovery)

Must study · Very likely · 100% · Medium

Asked in 4 of 4 papers · 47 marks in total

STUDY THESE

- VRF definition and advantages
- Three-pipe heat recovery sketch (liquid, suction, discharge lines + BC box)
- Two-pipe heat recovery block diagram; multi-split vs VRF

2024 Q7(b) 12 marks

What is meant by VRF system? Explain three pipe heat recovery VRF systems with necessary sketches.

2023 Q7(b) 7 marks

Distinguish between multi-split system and VRF system.

2021 Q7(a) 12 marks

What is meant by VRF system? Explain three-pipe heat recovery VRF systems with required sketches.

2021 Q7(b) 8 marks

What are the differences between a heat pump multisplit system and a VRF multisplit system?

2020 Q5(d) 8 marks

What are the advantages of VRF system? Draw the block diagram of two-pipe heat recovery VRF system.

Full answer: ESE 4025 Study Pack, topic B7.

B8 Cooling load: components, assumptions, numerical

Must study · Very likely · 100% · Hard

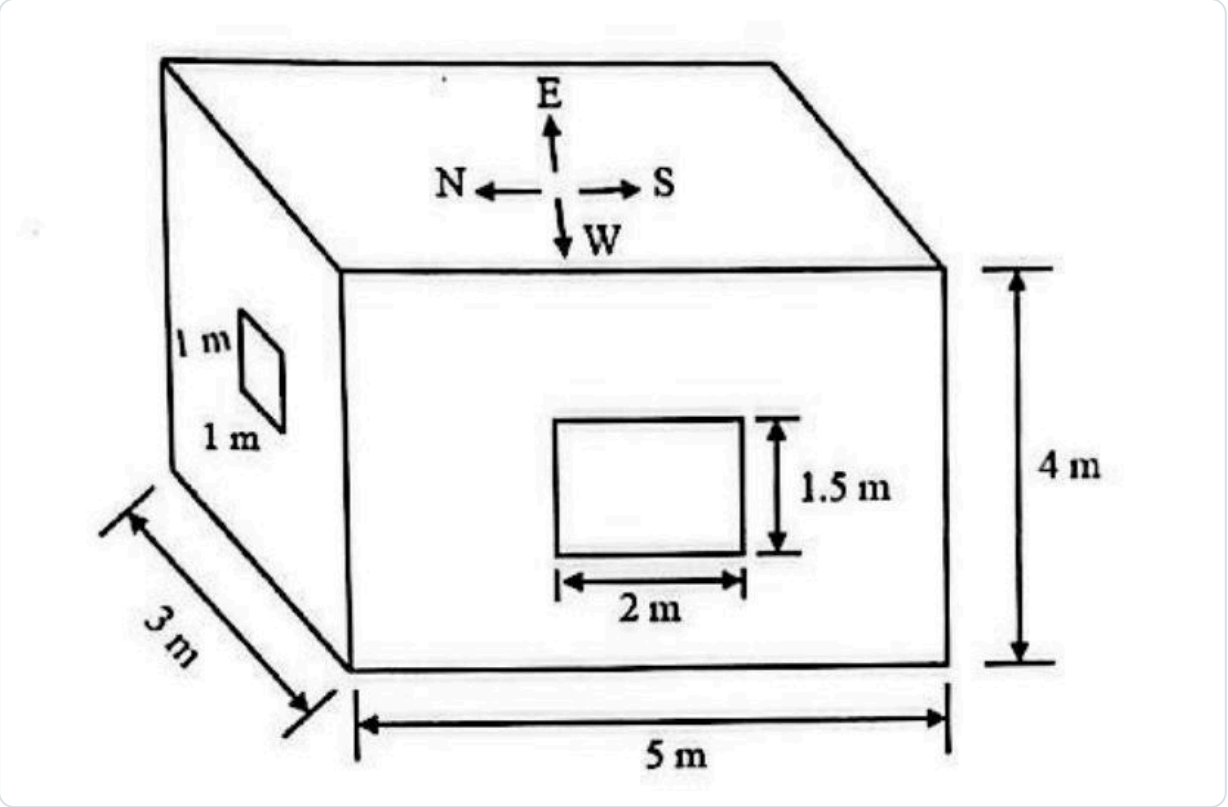
Asked in 4 of 4 papers · 54 marks in total

STUDY THESE

- Heating vs cooling load; components (walls, roof, glass, people, lights, appliances, infiltration, ventilation)
- Assumptions behind cooling load design
- Q = UA·ETD (wall/roof), glass = A × SHG × SC, people, lights, infiltration via ACH; grand total

2024 Q7(c) 15 marks

Two of the wall faces N and W contain double glaze window of a room as shown in figure. If there is no heat gain through other walls, what is the required cooling capacity of the room? Inside: DBT 43 °C, RH 55% (as printed). Outside: DBT 43 °C, WBT 24 °C. U-value of wall, roof and floor 1.78, 1.3, 1.2. ETD wall 25 °C, roof 30 °C. SHG 300 W/m^2, SC 0.86. Light 33 W/m^2 floor. Occupancy 4 (90 W sensible + 40 W latent each). Appliances 600 W sensible + 400 W latent. Infiltration 0.5 ACH.



2023 Q7(c) 7 marks

What are assumptions should be considered while calculating the cooling load?

2023 Q7(d) 15 marks

Define heating and cooling load. Describe how the cooling load in the form of grand total load on air conditioning system can be evaluated.

2021 Q7(c) 7 marks

What is meant by cooling load? What assumptions behind cooling load design?

2020 Q6(a) 10 marks

What is cooling load? List the main components of cooling load estimation.

Full answer: ESE 4025 Study Pack, topic B8.

B3 Summer AC with ventilation air and bypass factor (numerical)

Must study · Very likely · 90% · Hard

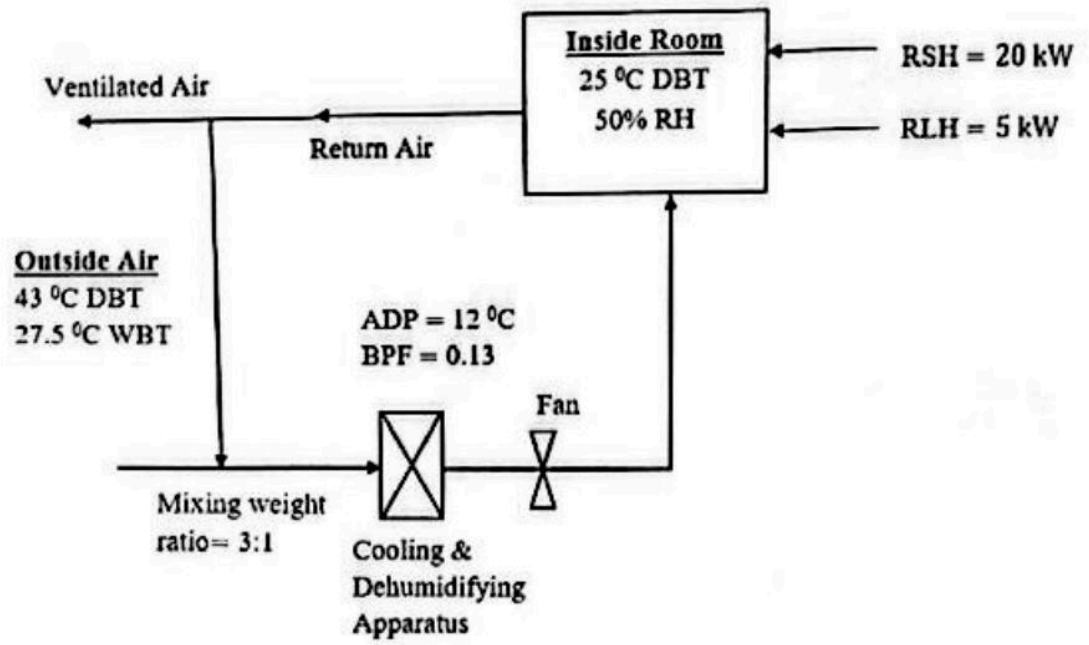
Asked in 3 of 4 papers · 41 marks in total

STUDY THESE

- Mix outside + return air (3:1), coil with ADP and BPF, supply to room
- RSHF line → ADP; dehumidified air = RSH / (0.0204 × (1−BF) × (t_r − ADP)) cmm
- OA sensible = 0.0204 × cmm × Δt; OA latent = 50 × cmm × ΔW; total load = sum / 3.5

2024 Q5(c) 15 marks

A refrigeration plant is shown in figure. Determine: (i) Dehumidified air quantity, (ii) Sensible and latent heat gain due to outside air, (iii) Total refrigeration load of the plant.



2023 Q5(d) 11 marks

Describe a summer air-conditioning with ventilation air-bypass factor.

2021 Q5(c) 15 marks

The room sensible heat gain and latent heat gain of a conditioned space are 40 kW and 15 kW respectively. The inside design conditions are 25 °C DBT and 55% RH. The return air is mixed with outside air before entering the cooling coil in the ratio of 3:1 by weight. Assuming the bypass factor of the cooling coil is 0.13, determine: (i) Apparatus dew point (ADP), (ii) Ventilation air in cmm, (iii) Total refrigeration load of the plant.

Full answer: ESE 4025 Study Pack, topic B3.

B10 Duct design rules, outlets, dampers, diffusers

Good to have · Likely · 60% · Easy

Asked in 3 of 4 papers · 35 marks in total

STUDY THESE

- General rules of duct design (short list)
- Factors for locating supply outlets and return inlets

- Functions of dampers and diffusers

> 2023 Q8(b) 7 marks
> What factors should be considered while selecting the location of supply air outlets?

> 2023 Q7(a) 6 marks
> Write down the functions of the following: (i) Dampers; (ii) Diffusers.

> 2021 Q8(c) 6 marks
> What are the general rules for duct design?

> 2021 Q8(b) 8 marks
> What factors should be considered to locating the supply air outlets and return air inlets?

> 2020 Q6(b) 8 marks
> What are the general rules of duct design?

Full answer: ESE 4025 Study Pack, topic B10.

B11 Sol-air temperature, window solar gain, infiltration

Good to have · Possible · 50% · Medium

Asked in 2 of 4 papers · 19 marks in total

STUDY THESE

- Sol-air temperature definition and formula
- Solar heat gain through glass = A × SHGF × SC × CLF
- Infiltration and the crack method

> 2023 Q8(a) 4 marks
> What is meant by sol-air temperature?

> 2021 Q7(d) 8 marks
> What is sol-air temperature? How to estimate the solar heat gain through building window?

> 2021 Q8(a) 7 marks
> What is infiltration? How to estimate infiltration rate using Crack method?

Full answer: ESE 4025 Study Pack, topic B11.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
ACH	air changes per hour
ADP	apparatus dew point
AHU	air handling unit
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
BPF	bypass factor
CAV	constant air volume
CFC	chlorofluorocarbon
CLF	cooling load factor
CLTD	cooling load temperature difference
COP	coefficient of performance
DBT	dry bulb temperature
DPT	dew point temperature
EER	energy efficiency ratio
ESHF	effective sensible heat factor
ETD	equivalent temperature difference
FCU	fan coil unit
GSHF	grand sensible heat factor
GTH	grand total heat
GWP	global warming potential
HCFC	hydrochlorofluorocarbon
HFC	hydrofluorocarbon
HVAC	heating, ventilation and air-conditioning
IAQ	indoor air quality
IIR	International Institute of Refrigeration
LiBr	lithium bromide

NH3	ammonia
OALH	outside air latent heat
OASH	outside air sensible heat
OATH	outside air total heat
ODP	ozone depletion potential
RE	refrigerating effect
RH	relative humidity
RLH	room latent heat
RSH	room sensible heat
RSHF	room sensible heat factor
RTH	room total heat
SC	shading coefficient
SEER	seasonal energy efficiency ratio
SHGF	solar heat gain factor
TEV	thermostatic expansion valve
TR	ton of refrigeration (3.5 kW)
TSH	total sensible heat
VAR	vapour absorption refrigeration
VAV	variable air volume
VCR	vapour compression refrigeration
VRF	variable refrigerant flow
WBT	wet bulb temperature

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